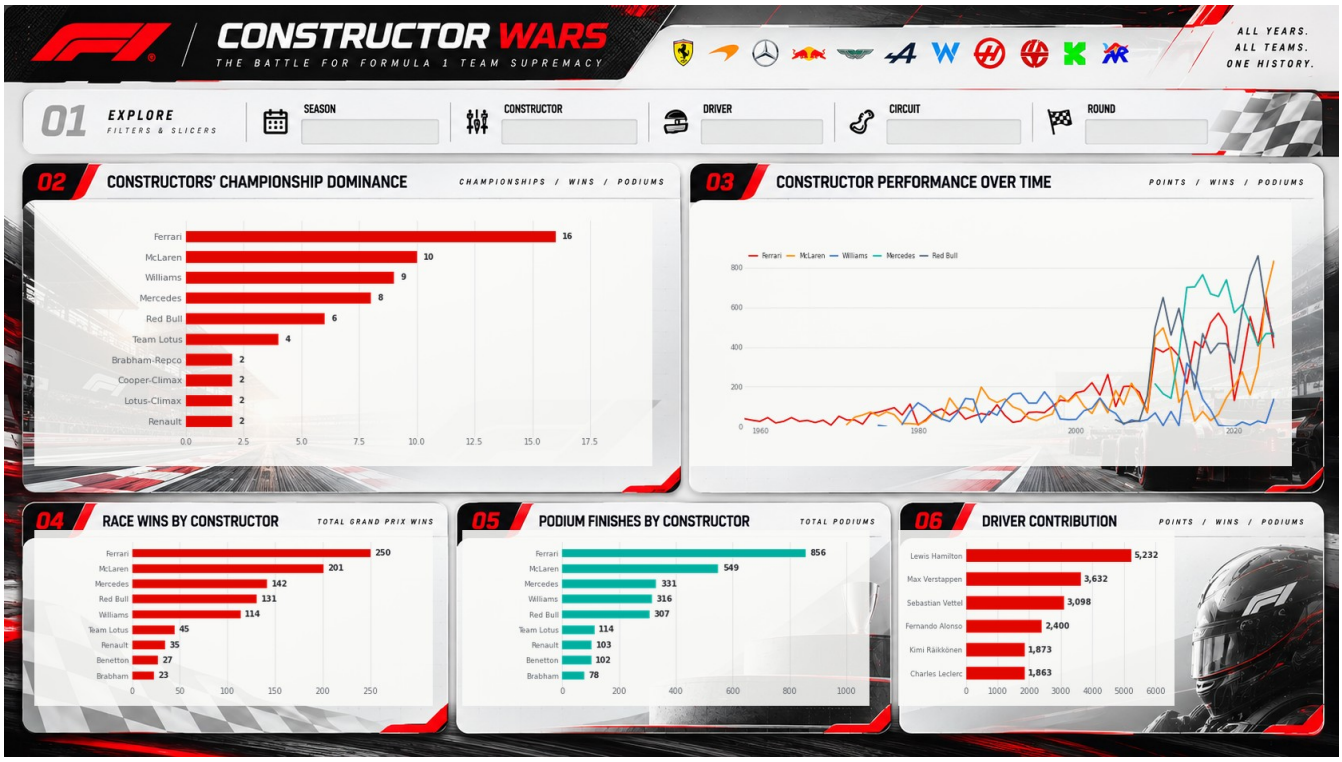
DATA VISUALISATION / PROJECT REPORT

CONSTRUCTOR WARS

THE BATTLE FOR FORMULA 1 TEAM SUPREMACY

Historical constructor performance, championship dominance, Grand Prix results and driver contribution



Static dashboard overview composed from the supplied HTML template and its embedded data. Filter controls in the original HTML remain interactive; this image is a report snapshot.

28,180	1,165	68	1950–2026
Embedded result records	Grand Prix events	Completed title seasons	Race-level coverage

Prepared from the user-provided HTML dashboard

Report date: 11 October 2026 | Analysis basis: in-file data and logic only

SOURCE BASIS: supplied HTML file and its embedded dataset

01 / AT A GLANCE

Executive summary

Constructor Wars is a single-file HTML dashboard that explores Formula 1 history through constructor championships, race wins, podium finishes, season-level points and driver contribution. The supplied HTML includes the underlying records and the JavaScript used to filter and aggregate them, so the analysis below follows those embedded rules rather than substituting external data.

16	250	856	5,232.5
Ferrari constructor titles	Ferrari GP wins	Ferrari GP podium finishes	Hamilton driver points*

Findings supported by the embedded data

- Ferrari leads the dashboard's all-time constructor-title count with 16 titles in the 68 completed championship seasons represented for 1958–2025. McLaren has 10, Williams 9, Mercedes 8 and Red Bull 6.
- Ferrari also leads the embedded constructor GP-win leaderboard (250), ahead of McLaren (201), Mercedes (142), Red Bull (131) and Williams (114).
- Ferrari leads constructor podium finishes (856), followed by McLaren (549), Mercedes (331), Williams (316) and Red Bull (307). Podium counts are driver-car appearances and preserve shared places where represented in the source records.
- Lewis Hamilton ranks first in the dashboard's driver-point total at 5,232.5. This metric sums points from both Grand Prix and sprint session rows; it is not normalized for historical scoring rules or season length.
- The latest completed season in the season-standings table is 2025. The HTML explicitly treats 2026 as incomplete and keeps it available for race-level analysis rather than completed-season title totals.

How to read the report

The figures use exact constructor labels present in the HTML, not a reconstructed corporate lineage. Historical entries such as "Team Lotus", "Lotus-Climax" and "Lotus-Ford" remain separate categories. Points and points-per-season are descriptive totals; changes in scoring rules over time mean that raw points should not be treated as a perfectly like-for-like performance measure.

> **Important source boundary**
>
> The supplied HTML does not expose a visible source URL, dataset version, extraction date, or full citation block in its embedded metadata. This report describes the embedded records and calculations; it does not claim that the raw data has been independently validated against an external source.

** Driver points follow the HTML's point aggregation across GP + sprint rows; other driver leaderboard metrics are Grand Prix-only.*

02 / DATA & MECHANICS

Dashboard architecture and method

Embedded data profile

Profile field	Count / range	Meaning in the HTML
Result records	28,180	All session-level records embedded in the HTML
Grand Prix rows	27,590	Session type code 0 in the dashboard logic
Sprint rows	590	Separate sprint-session rows
Grand Prix events	1,165	Unique Grand Prix event IDs represented
Year coverage	1950–2026	Race-level records; 2026 is incomplete
Constructors / drivers / circuits	213 / 865 / 78	Distinct indexed labels in the embedded dimensions
Season-constructor rows	930	Season standings rows supplied for constructor analysis
Completed title seasons	68	1958–2025 inclusive

Filters and chart controls

The dashboard exposes a season range, constructor multi-select, driver multi-select, circuit multi-select, round multi-select and a reset control. Three chart sections include metric toggles: championship dominance (titles / wins / podiums), constructor trend (season points / GP wins / GP podiums) and driver contribution (points / GP wins / GP podiums).

Aggregation rules implemented in the HTML

Metric	Source grain	Rule / limitation
Championship titles	Completed season table, 1958–2025	Count official title flags by constructor; driver, circuit and round filters do not change titles.
Season-final points	Constructor-season standings rows	Use final season points, not a sum of cumulative standings across event rounds; missing seasons are left blank.
Constructor GP wins	Grand Prix result rows only	Deduplicate by event + constructor so a shared driver win does not inflate team/event wins.
Constructor GP podiums	Grand Prix result rows only	Deduplicate by event + driver + constructor; podiums represent driver-car appearances.
Driver points	All session rows	Sum points over GP and sprint rows; raw historical point totals are not scoring-normalized.
Driver GP wins / podiums	Grand Prix result rows only	Count GP events per driver; sprint wins and podiums are not included in these modes.

SOURCE BASIS: supplied HTML file and its embedded dataset

03 / TITLES

Constructor championship dominance

The championship view counts official constructors' titles from the completed-season table. In the embedded records, 68 title seasons are represented from 1958 through 2025.

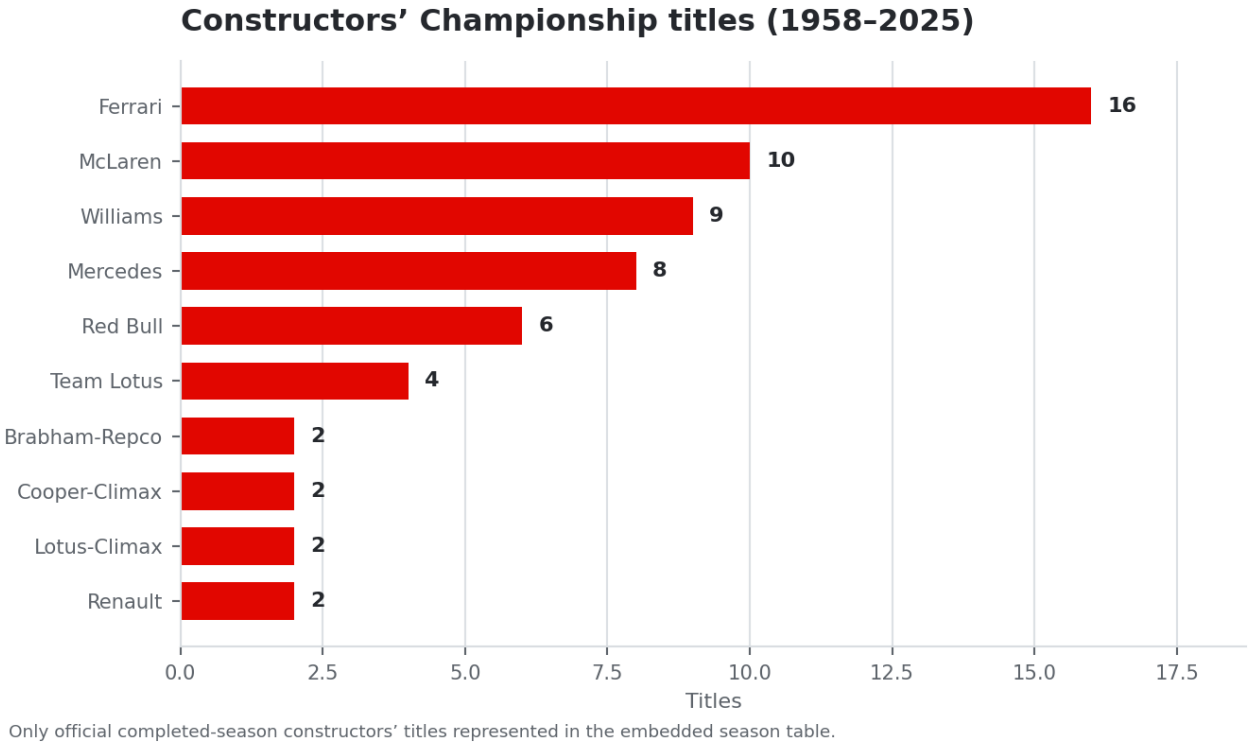


Figure 1. Constructor titles by exact embedded constructor label; sorted by title count.

Constructor label	Titles	Share of 68 seasons
Ferrari	16	23.5%
McLaren	10	14.7%
Williams	9	13.2%
Mercedes	8	11.8%
Red Bull	6	8.8%
Team Lotus	4	5.9%
Brabham-Repco	2	2.9%
Cooper-Climax	2	2.9%

What stands out

- Ferrari has 16 titles, equal to 23.5% of the 68 season titles represented in the table. McLaren has 10; Williams has 9; Mercedes has 8; Red Bull has 6.
- The five highest title-count labels account for 49 of 68 titles (72.1%). This indicates that the title leaderboard is concentrated among a small number of constructor labels, while the rest are distributed across historical entries.
- The label-level method means "Team Lotus", "Lotus-Climax" and "Lotus-Ford" appear separately. This is faithful to the in-file categories but not a consolidated view of Lotus lineage.

SOURCE BASIS: supplied HTML file and its embedded dataset

04 / RACE RESULTS

Grand Prix wins and podium finishes

Constructor race-result charts use Grand Prix rows only. They exclude sprint victories and podiums. Constructor wins are deduplicated at event + constructor grain; podiums are counted per driver + constructor + event.

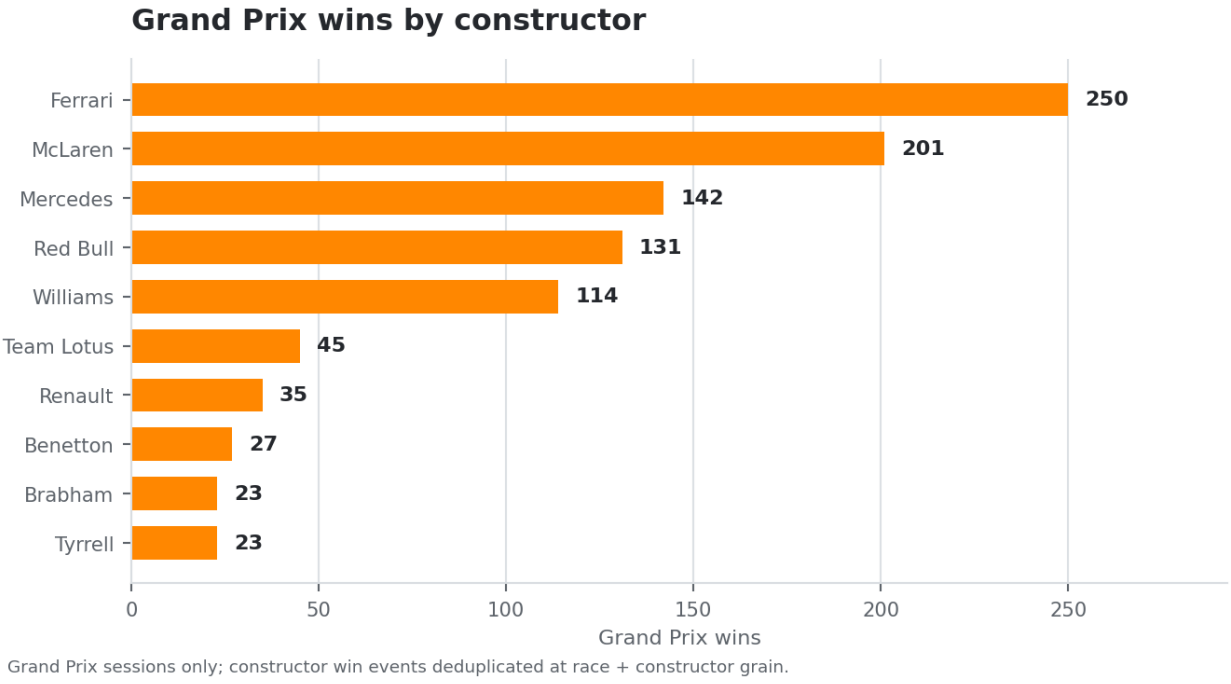


Figure 2. Grand Prix wins by constructor; race + constructor deduplication is applied.

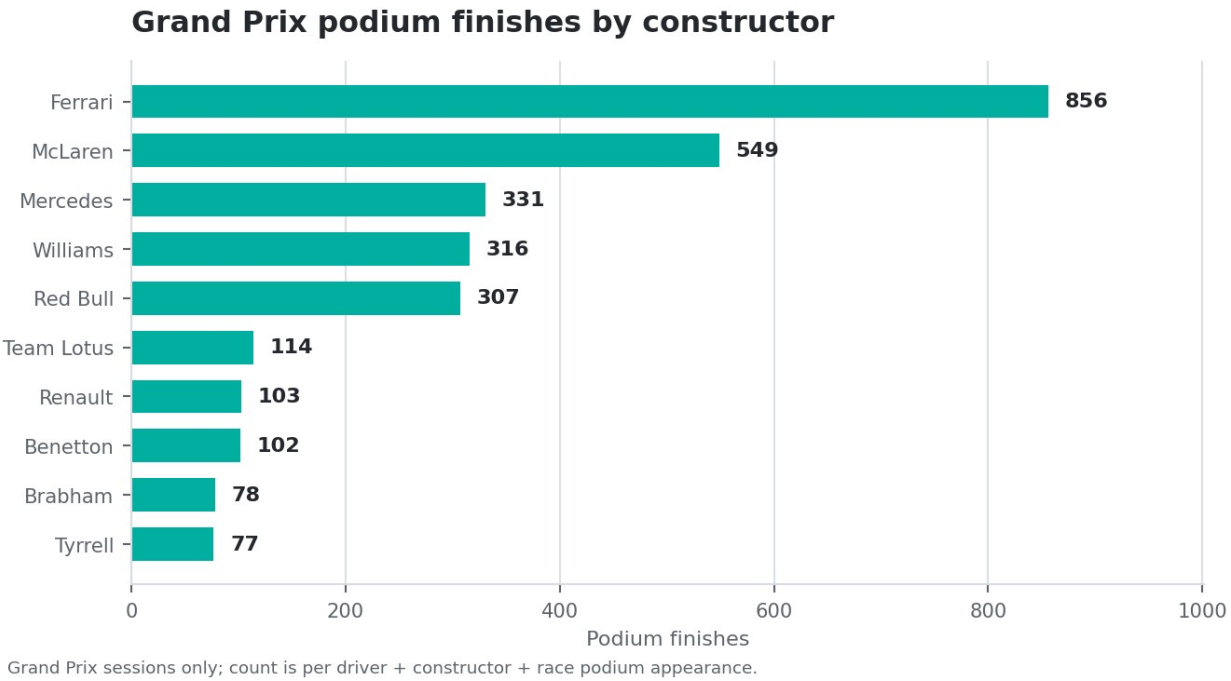


Figure 3. Grand Prix podium finishes by constructor; driver-car event podium appearances.

Comparison

Ferrari records 250 GP wins compared with McLaren's 201, a difference of 49. In podium finishes, Ferrari records 856 compared with McLaren's 549, a difference of 307. Mercedes has 142 wins and 331 podium finishes; Red Bull has 131 wins and 307 podium finishes; Williams has 114 wins and 316 podium finishes.

SOURCE BASIS: supplied HTML file and its embedded dataset

05 / SEASON TREND

Constructor performance over time

The default trend compares final season points for the five constructor labels with the highest title counts in the embedded completed-season table. The chart spans 1958–2025; only seasons with a constructor-season record are plotted for that label.

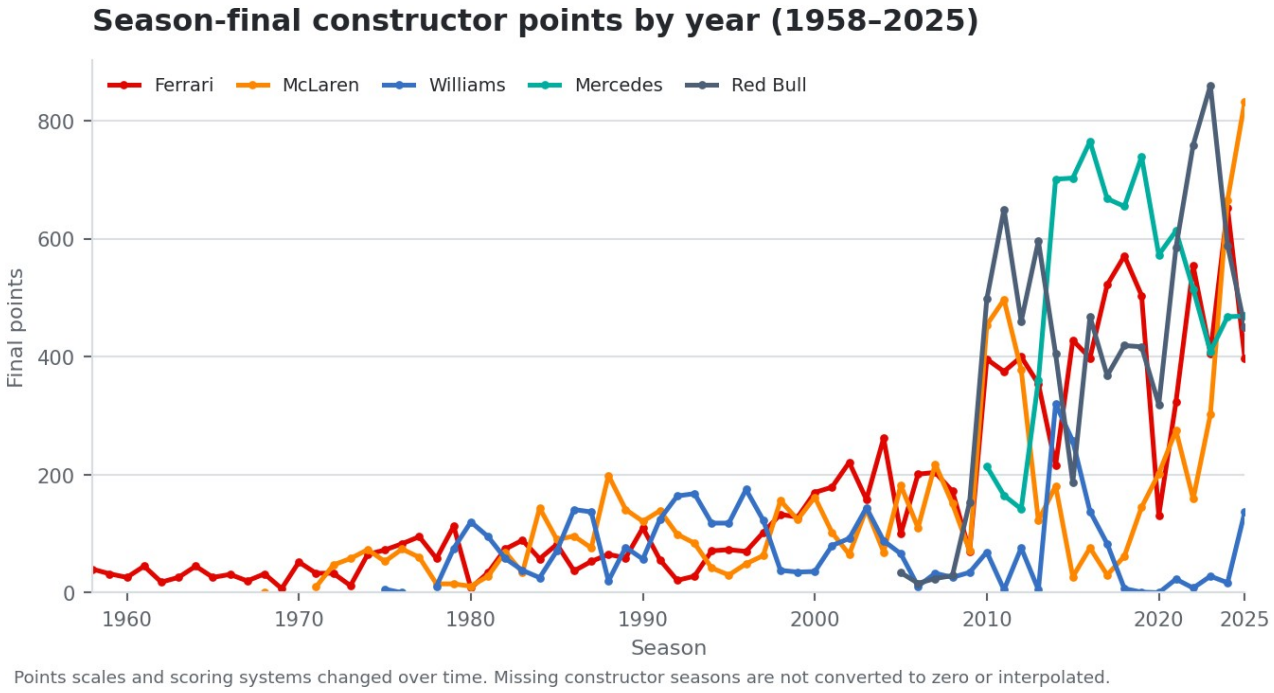


Figure 4. Season-final points, title-leading constructor labels; gaps represent years with no season row for that exact label.

Reading the curve responsibly

- The data show season-level totals, not race-by-race cumulative points. That matches the HTML's explicit rule that the season points line is sourced from final standings.
- Raw points grew and changed meaning as scoring systems changed. A taller modern peak does not by itself prove a stronger car or team than an earlier era.
- The HTML deliberately breaks a series when a constructor-season row is missing rather than filling the gap with zero or drawing an interpolated line. The report's graph preserves this behaviour.
- The last completed season in this summary is 2025. The HTML notes that 2026 is incomplete and available for race-level analysis only.

Dashboard interaction

The live HTML lets users switch this panel from season-final points to GP wins or GP podium finishes. For the race-result modes, the chart responds to constructor, driver, circuit, round and year filters; the season-final points mode is based on the standings table and is not altered by driver/circuit/round filters.

06 / DRIVER COMPARISON

Driver contribution

The driver contribution panel supports points, Grand Prix wins and Grand Prix podium finishes. The point view is deliberately broader than the win/podium views: it sums points in Grand Prix and sprint session rows; wins and podiums remain Grand Prix-only.

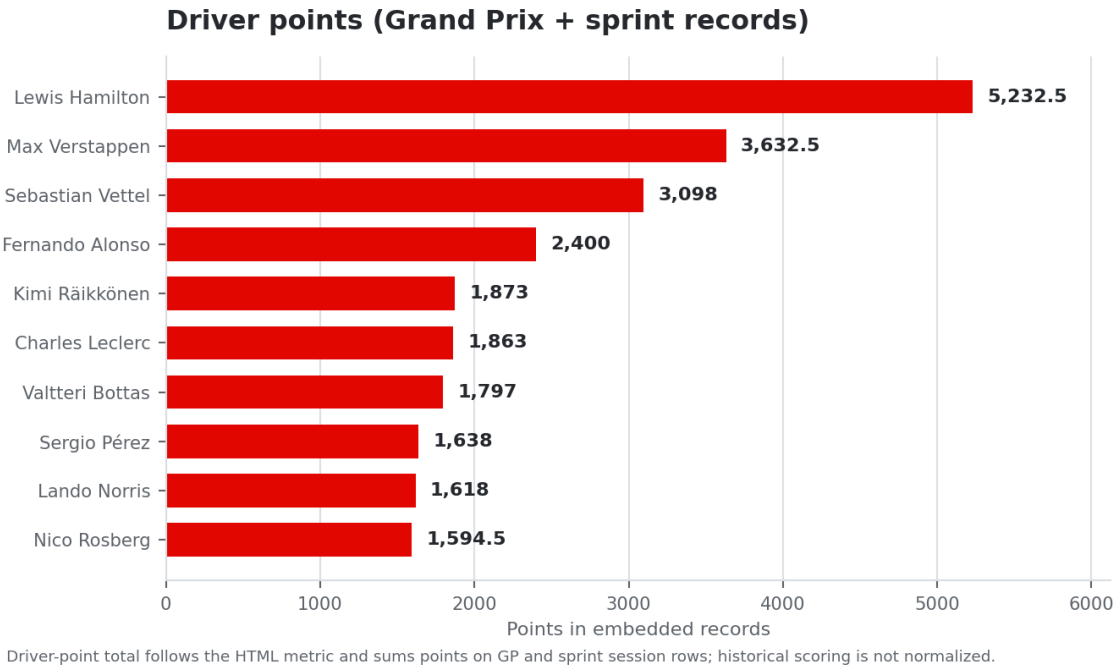


Figure 5. Driver points from the embedded GP + sprint session rows. Totals are not normalized across scoring eras.

Separate Grand Prix-only driver leaders

Rank	GP wins leader	Wins	Podium leader	Podiums
1	Lewis Hamilton	106	Lewis Hamilton	208
2	Michael Schumacher	91	Michael Schumacher	155
3	Max Verstappen	72	Max Verstappen	135
4	Sebastian Vettel	53	Sebastian Vettel	122
5	Alain Prost	51	Alain Prost	106
6	Ayrton Senna	41	Fernando Alonso	106
7	Fernando Alonso	32	Kimi Räikkönen	103
8	Nigel Mansell	31	Ayrton Senna	80
9	Jackie Stewart	27	Rubens Barrichello	68
10	Jim Clark	25	Valtteri Bottas	67

Interpretation

- Lewis Hamilton has the highest total in all three driver summaries constructed from the embedded rows: 5,232.5 points (GP + sprint), 106 GP wins and 208 GP podium finishes.
- Points and win/podium rankings answer different questions: points accumulate across finishes, while wins and podium counts track specific Grand Prix results.
- Points should not be used as a simple cross-era skill ranking: they depend on scoring rules, eligible events, sprint availability and the length of each driver's career in the dataset.

SOURCE BASIS: supplied HTML file and its embedded dataset

07 / RESPONSIBLE USE

Limitations, validation and conclusion

Limitations visible from the supplied HTML

Limitation	Implication
Source provenance is incomplete	No visible citation URL, version identifier, extraction timestamp or full source manifest appears in the supplied HTML metadata. The dataset origin needs to be documented before external submission.
Historical points are not normalized	Scoring systems, calendars and sprint formats differ across eras. Points are descriptive totals, not directly equivalent performance units.
Constructor labels are not consolidated	Historical labels remain separate. A brand/lineage comparison would require an explicitly documented alias map and transparent treatment of engine/entrant names.
2026 is unfinished	The HTML states that 2026 is available for race-level analysis, while season-final constructor standings are represented only through 2025.
Embedded row schema is compact	The dashboard uses numeric array positions and indexed dimension arrays. A publishable dataset should include a readable data dictionary for each field and session-type code.
No independent source validation here	This report recalculates the summaries from the HTML's embedded data and in-file logic; it does not assert that the original data has been cross-checked against an outside authority.

Recommended next steps

- Add a source-and-lineage note to the dashboard: provider URL, download/version, retrieval date, licensing/attribution requirements and update cadence.
- Publish a data dictionary for the embedded record arrays, including the meaning of every index, the GP/sprint type code, race ID grain, event result flags, and season-table columns.
- Offer two clearly labelled comparison modes: "raw historical totals" and, where methodologically defensible, normalized metrics such as points per event or within-season rank. Do not blend them silently.
- Provide an optional alias grouping for historical constructor families while preserving the raw constructor label so users can reproduce the original rankings.
- Add tests for title season counts, duplicated winner rows, shared podiums, active filters, zero-match states, year-range boundaries and 2026 incompleteness.

Conclusion

The uploaded dashboard already combines an accessible historical scope, useful filter controls and metric-specific aggregation rules. Based on its embedded data, Ferrari leads the title, constructor GP-win and constructor podium leaderboards; McLaren ranks second in titles and GP wins; and Lewis Hamilton tops the driver totals. The dashboard is most useful when users distinguish championship outcomes, GP event results and accumulated points rather than treating them as interchangeable indicators.

> **Report scope**
>
> All numeric findings and chart values in this report were computed from the embedded JSON data and aggregation rules found in the supplied HTML file. No outside facts or dataset source claims were added.

Reference artifact: Pradip_Lalpura_ADS_BATCH-4_DATA_Visualisation_constructor_wars_dashboard.html

SOURCE BASIS: supplied HTML file and its embedded dataset